RB GenAI Assignment:
Evaluating Generative Agent Simulation Performance

Project Briefing

You are hired by a Beauty & Wellbeing company, a global leader in skincare, haircare, and personal care products. The company offers a diverse portfolio of brands catering to a wide range of consumer needs, from premium skincare and professional haircare to everyday beauty essentials. These products are distributed through various channels, including standalone brand stores, major retail chains, and e-commerce platforms. One of the firm's primary goals is to increase sales with young consumers, specifically targeting business professionals who work long hours and are frequent travelers.

The company has adopted a GenAI approach similar to the paper *"Generative Agent Simulations of 1,000 People"* (https://arxiv.org/pdf/2411.10109), with the goal to create *human simulations* based on in-depth interviews with their target demographic: young business professionals with demanding lifestyles. A *human simulation* is an LLM mimicking the traits and behaviors of a real person. Two interviews, containing only open-ended questions, were conducted per person:

1. The first set of questions was used to generate the *human simulations*.
2. The second set of questions was designed for evaluation, and you will have access to a selected subset of these questions in this test.

The interviews were conducted via a text-based chat interface.

Assignment

Now, the company wants you to assess how accurately these human simulations capture the nuances of individual human behavior of their target demographic. This assessment should be based on a comparison between the responses of the simulated humans and the real humans from the second interview.

Your task is to evaluate how accurately the human simulations' responses mimic real human answers to the open-ended questions from the second interview. Attached you find a dataset comprising paired responses to these market research questions - one set from real human participants and the corresponding set generated by their *human simulations*.

You may consider the following:

- Define specific evaluation criteria, detailing what to evaluate, measurement methods, and the reasons for these choices.
- Measure how closely the agents' responses and interactions align with real human behavior in the given contexts.
- Identify strengths and weaknesses of your approach and suggest potential improvements.
- Highlight areas where the *human simulation* does not fully mimic human responses.
- Take time to reflect on the approach and critical aspects of your evaluation before you begin.

Approach: You are free to use any GenAI-based approach for the evaluation. You may choose a single method or multiple approaches (e.g., LLM chains, graph-based agents, decoder/encoder-based transformer models, or any other GenAI approach that comes to mind).

Final Deliverables

- **Code** (if applicable)**:** Final source code of your solution (please follow common coding guidelines).
- **Technical Report:** A report (max. 2 PDF pages) detailing your evaluation concept, results, and other relevant insights and implications. The format and design of the report is up to you.
- **Pitch Slide Deck:** A slide deck designed to convince the CEO and the head of consumer insights that this evaluation method is the right approach for assessing the firm's human simulations. The CEO is used to slide decks in the style of leading strategy consultancies (regarding level of clarity, structure, visuals and professionalism). Limit the deck to three core slides (excluding introductory and any supplemental slides).

Submission Deadline: Please sent the deliverables within **24 hours** of receiving this assignment email. We recommend completing the work within **8 hours.**
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Data Set

Please use the dataset "RB_GenAI_Datatest.xlsx" (which only contains a subset of the original dataset). This file contains both **human** and ***human simulation*** responses from 3 different humans on 10 different open-ended questions.

The dataset is structured as follows:

- **Question Category:** The market research category the question belongs to (e.g., shopping behavior and product preferences).
- **Question Text:** The open-ended question itself.
- **Person ID**: The unique identifier of the human. There are only 3 humans included in the subset, who were interviewed to generate their *human simulation.*
- **Human Answer:** The response provided by the *human consumer***.**
- **AI Answer:** The response generated by its *human simulation*.

Note: This test is purely artificial and has no connection to real companies.